

Generating Synthetic Earthquake Response Spectra from an Interpretable Hierarchical Latent Space

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Abstract

Generative ground motion models are a key component in deep learning-based earthquake early warning systems, used to predict the future development of ground motion from initial seismic tremors. However, the physical attributes of the datasets used in such models are generally not utilised to their full extent. This project proposes a hierarchical variational autoencoder for generating synthetic earthquake response spectra from a highly interpretable latent space. The model structures a latent space in Poincaré ball geometry and uses a two-tier hierarchy to regularise embeddings with respect to both event-level features such as magnitude and sample-level features such as rupture distance. The model achieves strong reconstruction accuracy with a mean R^2 of 0.969, above 0.92 for all spectral periods. The latent variables have clear correlations with magnitude and rupture distance, Pearson correlation coefficients of -0.844 and 0.615 respectively, indicating the greater interpretability. Some example synthetic response spectra were subsequently generated from the trained model with a two-stage neural network approach to estimate associated feature values from points in the latent space. Finally, a feed-forward neural network is used to predict a point in the latent space from live intensity measures, demonstrating a practical application within early warning systems. Further work could consider more complex hierarchical structures, perhaps including further tiers or grouping samples by station rather than event. This project highlights the positive impact of incorporating physically meaningful attributes of data into generative ground motion models.

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1 Introduction

1.1 Background

Earthquakes are among the most devastating natural disasters, accounting for nearly 60% of all natural disasters fatalities, including hurricanes and floods [1]. Between 1998 and 2017 alone, earthquakes caused approximately 750,000 deaths and displaced or injured over 125 million people worldwide [2], highlighting their significant global impact. Various strategies exist to mitigate the damage and improve preparedness. For example, earthquake early warning systems (EEWSs) predict the future development of an earthquake minutes before it occurs from a few seconds of initial seismic tremors, offering critical seconds to minutes of warning for evacuation or safety measures [3, 4]. Some approaches to EEWS embed earthquake recordings into a latent space for generative modelling [5, 6]. The structure of this latent space is a crucial component in the performance and interpretability of such frameworks, and improving the design could lead to more robust and transparent EEWS.

While EEWS provide valuable early warnings, earthquake events themselves are often inevitable given common statistics and the inherent unpredictability of geophysical processes [7]. Consequently, another critical mitigation strategy lies in designing buildings to minimise damage when earthquakes do occur. With modern computing power and software packages, accurate building simulations can be used to model such effects [8]. In particular, consider a building represented as a multiple degree-of-freedom (DOF) system, where each DOF represents a singular mode of vibration for some part of the building [9]. Each DOF has some associated natural period at which the part of the building would maximally oscillate. By considering the acceleration response spectrum of a given earthquake, one can model the force exerted on each DOF to predict the impact that a particular earthquake would have on the building. However, earthquake datasets are inherently limited due to the unique and complex nature of each event and recording. This scarcity poses a challenge for simulation-based design. As such, the capability to enrich earthquake data through synthetic generation can be hugely beneficial for building design simulations and wider ground motion simulation applications too.

The response spectrum of an earthquake is a fundamental concept in seismology, representing the intensity of an earthquake at a specific site as a function of structural period, commonly referred to as the *period domain*. Consider an object with natural period T . The response spectrum $S_a(T)$ represents the maximum acceleration experienced by that object throughout the duration of the earthquake. In this form, it is intuitive to compare earthquakes to one another and infer broad insights such as magnitude, rupture distance, and local soil conditions. It is important to note as well the methodologies taken to ensure accuracy and consistency in the recording of response spectra. An approach to recording ground motion by Boore et al. (2006) [10] utilises orientation-independent measures to avoid bias introduced by arbitrary sensor alignment, ensuring that response spectra reflect true ground motion rather than artefacts of measurement direction.

Deep learning (DL), and more generally machine learning (ML), approaches have seen an exciting level of success and interest over the past few years in a range of different domains [11], and the applications of such approaches within the context of earthquake mitigation strategies has been highly effective [12, 13]. Advances in the field of ML has lead to a huge uptake in applications to seismology, offering data-driven approaches to statistical analysis and modelling. EEWS, and more broadly earthquake detection, is a crucial topic within seismic modelling, utilising models to flag and classify earthquakes from intensity measures (IM), often in real-time [5, 14, 15]. Convolutional neural networks (CNNs) and recurrent neural networks (RNNs) have significantly improved the sensitivity and speed of detecting seismic events from continuous waveform data [16, 17, 18]. Similarly, ML techniques have also seen great success within earthquake localisation [19], determining the epicentre and hypocentre of events is essential for timely and effective hazard response. One of the main challenges in the application of ML approaches is their interpretability. Some research has focussed on improving interpretability of models within some domains and seen great success [6, 20, 21], this aspect should not be overlooked.

1.2 Research Gap & Project Goals

Previous research into generative earthquake modelling has primarily centred around producing realistic samples, exploring the different representations of earthquakes and model frameworks which work best for this domain. Although some interest has been seen with the interpretability of the latent space used for these generative models, certain aspects of the data have yet to be considered. Latent space interpretability is highly important when analysing relationships between embedded samples and considering applications of these models. If similarly accurate data can be generated from a latent space encoding a higher level of meaning, then the applications within broader frameworks such as EEWS would be valuable.

The primary focus of this project is on incorporating the relationship between different samples of the same event to create a latent structure for synthetic response spectra generation regularised at two levels to improve interpretability. More specifically, this project proposes a hierarchical prior distribution within a variational autoencoder (VAE) framework, borrowing concepts from mixed effects modelling, to create a partial pooling effect where embedded samples are grouped by event. The first level of the hierarchy encodes variation between different events, regularised and structured using event-level features such as magnitude. The second level of the hierarchy encodes spatial correlation between different recordings of the same event by regularising the embeddings with sample-level features such as rupture distance. In effect, one can consider sampling a point in the trained latent space as defining the centroid of an earthquake event, then taking further samples surrounding this centroid as corresponding recordings of the event from different stations. Figure 1 demonstrates this approach more visually.

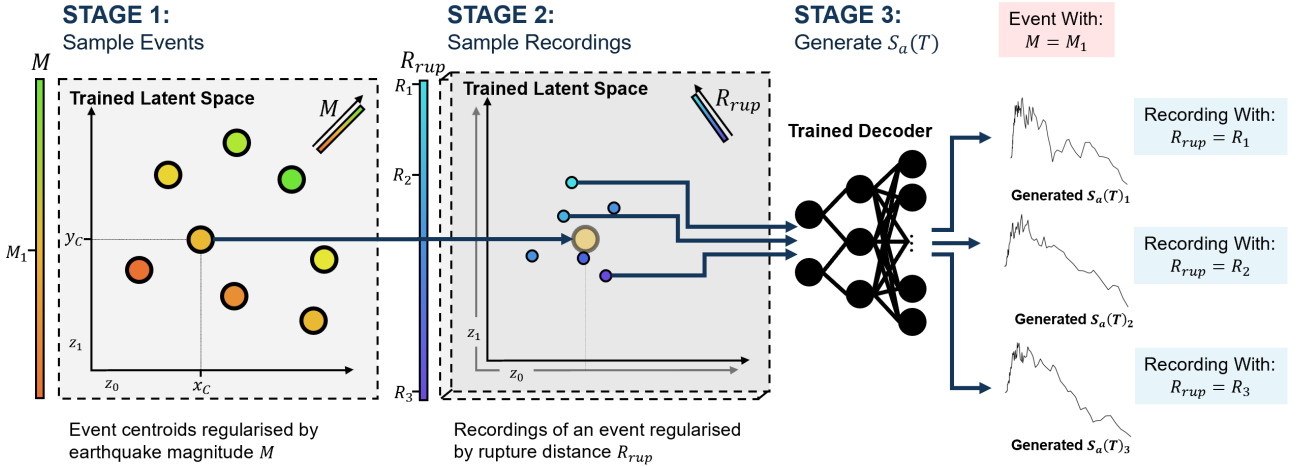


Figure 1: Stage 1 involves sampling an event centroid from the latent space (x_C, y_C) , noting estimated event-level features such as magnitude $M = M_1$. Stage 2 involves sampling a recording in the latent space surrounding (x_C, y_C) , noting estimated sample-level features such as rupture distance $R_{rup} = R_1$. Stage 3 involves passing the latent space coordinates of the recording sample to the decoder, to generate a synthetic response spectrum such as $S_a(T)_1$. Multiple samples can be taken surrounding (x_C, y_C) to simulate recordings from multiple stations.

In practice, the proposed generative model could be implemented within a broader EEWS framework using a regression approach to predict a point in the latent space from initial seismic tremors which can subsequently be decoded into response spectra with associated feature values. The higher degree of interpretability in this generative model enables better estimation of feature values which can aid the reconstruction of the time-domain representation from the response spectra.

2 Related work

2.1 Seismic Applications of Machine Learning

A recent paper by Fayaz et al. (2025) [6], highly influential for this project, proposed an application of DL for a domain-informed EEWS. The proposal contained three key components: a domain-informed VAE, Gaussian process regression (GPR), and a feed-forward neural network (FFNN). The VAE learns latent embeddings of earthquake recordings from response spectra, while the FFNN learns a mapping from live IMs to this latent space, allowing decoding into response spectra. GPR estimates response spectra at additional locations, and Bayesian updating refines these estimates as seismic waves arrive at different sites, enhancing predictions. The VAE incorporated further regularisation of the latent space by considering the magnitude and epicentre distance relevant to each recording, keeping similar samples embedded closely. This novel approach enables real-time on-site predictions from a meaningful latent space whilst incorporating information from different stations recording the same event at test time. The VAE aspect of the research offered a solid foundation for the model developed in this project, but further steps could be taken towards creating a generative model considering more physical aspects of the data to improve interpretability.

Over the past year, there has been several interesting papers published on generative ground motion models (GMMs) [16, 22, 23, 24], exploring a range of different ML techniques. In particular, Chen et al (2024) [16] saw significant success utilising a generative adversarial network (GAN) with convolutional layers to model the seismic waveform from the Shandong Pingyuan earthquake of 2023. Their approach involved a conditional structure in the latent space by feeding features such as earthquake magnitude to the generator network. Although the approach was successful, the full generalisability of it remains a question since the research only used samples from one singular event.

Since earthquakes are real physical events, the decision on how these events are processed computationally is an important one which must be considered carefully. For example, using the acceleration response spectrum is indeed very common and the approach that shall be taken in this project, however it is interesting to note other options which have been investigated previously. A recent paper by Ning et al. (2024) [23] explored a generative GMM via a convolutional VAE with the acceleration spectrogram of an earthquake. The spectrogram can be thought of as a 2D representation of acceleration detailing time variance along one dimension and frequency variance along the other. Computed via the short term Fourier transform of the seismic waveform, it effectively shows the progression of the acceleration response spectrum over time. The spectrogram saw further success with Bergmeister et al. (2024) [24] through a de-noising diffusion-based model. The approach in this paper involved a VAE model to downsample the spectrogram before feeding the downsampled latent embeddings to the diffusion model. This two stage approach was primarily taken due to the complexity and high dimensionality of the acceleration spectrogram. This perhaps is one of the main drawbacks to this representation; although it captures a high degree of complexity of the earthquake, the intricacies can be challenging to model.

While generative GMMs are often used as a component within DL-based EEWS frameworks, the synthetic generated data from these models can also be utilised more explicitly to improve the performance of other predictive earthquake models. Monterrubio-Velasco et al. (2024) [25] demonstrated this by implementing an ML estimator for real-time earthquake predictions trained entirely on synthetic ground motion data. The use of synthetic data can effectively bypass the limitations of data sparsity and regional variability to enhance predictions at a particular site. In contrast, Wang et al. (2025) [26] explored a generalisable DL approach for the prediction of laboratory earthquakes. Training samples are carefully created under laboratory conditions with clear fault slip mechanisms and acoustic recordings producing accurate recorded data, mimicking seismic behaviour. Their approach trained a deep CNN on one set of laboratory conditions and demonstrated transfer learning via fine-tuning to predict time-to-failure and shear stress in different laboratory conditions. These papers highlight the capability of training models on highly controlled data, such as synthetic or laboratory data, for improved predictions. There is clear interest in finding approaches to extend ground motion

datasets for better predictions on unseen samples, making synthetic data a primary resource within data-driven seismology.

2.2 Generative Modelling

Generative modelling is an exciting area of ML which has gained considerable traction in recent years [27], particularly with the development of more powerful large language models such as ChatGPT [28]. The gradual progression from early implementations of the VAE (2013) [29] to the development of the GAN (2014) [30] and diffusion-based models (2015) [31] has been a large driving force behind the adoption of generative ML to a range of domains. These three approaches represent the primary frameworks behind most current generative models [32] and all have benefits and drawbacks given their considerable differences. For example, VAEs learn an embedding of samples, regularised by some KL-divergence, which can be decoded to generate realistic samples. In contrast, GAN models typically learn to generate synthetic samples from some random latent space which are improved using an adversarial loss by training a discriminator to distinguish synthetic and real samples. Finally, diffusion-based models recursively add noise to a sample and learn a mapping to remove the noise at each stage, resulting in a model which can generate high quality synthetic samples from random white noise. The main benefit to both the GAN and diffusion-based models over the VAE is the higher quality of generated samples, VAEs typically produce "blurry" results [32]. However, the VAE framework still has merit given the inherently well-structured latent space which can easily be made highly interpretable through regularisation techniques. In general, despite being older, the VAE is still widely used in current research for this reason.

Capturing hierarchical relationships within ML models is an interesting challenge with many significant benefits. A common approach to this is mixed effects modelling [33, 34, 35, 36] which introduces a partial-pooling effect between groups in the hierarchy. For example, consider samples of earthquakes to be representative of a particular event at a particular station, each sample could hence be grouped by either event or station to create a two-tier hierarchy. Mixed effects modelling typically works by considering parameters of a model to be some probability distribution which is sampled from for each group; in other words the parameters are conditioned on the groups. Generally, this is implemented by defining the parameter as a sum of two components: the *fixed effect* and the *random effect*. Each sample is processed by the model using a universal value for the fixed effect (determined by the entire population of samples) and a random effect relevant to the corresponding group (determined only by samples in the group). This means that predictions are dependent on the entire dataset whilst being tailored towards each distinct group.

Traditionally, mixed effects modelling is applied to regression tasks, but applications within generative modelling have also been investigated [35, 36]. A recent paper by Simchoni et al. (2024) [35] proposed a direct integration of mixed effects modelling into the VAE framework. Their approach used two separate encoders: one to capture fixed effects and another to model random effects. The random effect encoder produces embeddings in the same space as the input features, which are then added to the reconstruction generated from the decoded fixed effect embedding. This additive structure mirrors traditional linear mixed effects models and enables the VAE to capture complex dependencies, such as hierarchies, between different samples in the data. This approach is quite computationally heavy particularly if there are a large number of discrete groups in the data. In contrast, Ong et al. (2024) [35] demonstrated success applying mixed effects modelling to VAEs by making the prior distribution used in the KL-divergence loss term linearly dependent on group level features. This simpler approach is much easier to implement, however the main danger is that, if two distinct groups share the same group level features, they will be inadvertently joined together. Aspects of this approach, such as representing the prior distribution as a mixed effects variable, were highly influential to this project.

When working with a VAE model, desired structure of the latent space can be attained in a variety of ways, for example through the introduction of regularisation loss terms [6]. Another consideration to make is the intrinsic geometry of the latent space, approaches can be taken which add curvature to encourage the space to take a particular shape or allow for a desired structure to fit more naturally.

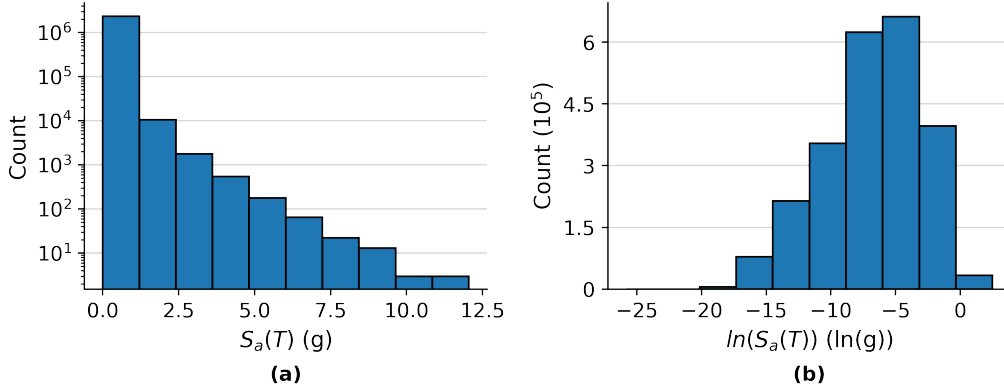


Figure 2: Histogram of the acceleration response spectrum values from the dataset, (a) raw values ($S_a(T)$), (b) log-transformed ($\ln S_a(T)$).

This adaptation is known as *hyperbolic geometry* and has been studied as a possible application within the VAE framework [37, 38]. There are many ways to incorporate hyperbolic geometry, a common approach being the Poincaré ball such as in the Poincaré-VAE model proposed by Mathieu et al. (2019) [38]. Their approach models the space as a hyper-ball where the edge of the ball is considered to be infinitely far from the origin. In essence, distances grow exponentially away from the origin which is beneficial to hierarchical structures where deeper levels of the hierarchy require more intricate detail. Such geometry was used in this project in an attempt to better capture the relationships between grouped samples in the latent space.

3 Methodology

The primary objective of this project is to develop a model framework for generating synthetic earthquake response spectra from a highly interpretable latent space. This section outlines the methodology with step by step details, covering the dataset used, the base framework implementation, and the subsequent modifications made to achieve the intended outcomes.

3.1 Dataset

The NGA-West2 ground motion dataset comprises 21,540 samples of 600 earthquakes recorded from 3,039 distinct stations [39]. The features of the dataset primarily fall into one of three categories: event-level features (describing the earthquake), sample-level features (describing the site of recording or recording details relevant to the event), and IMs specific to each sample. As discussed, the most critical IM for this project is the acceleration response spectrum, $S_a(T)$. Accordingly, the characteristics and distribution of this feature are explored first.

Figure 2a shows the distribution of $S_a(T)$ values across the dataset, plotted on a logarithmic scale for improved visibility. As expected, the distribution is heavily skewed toward lower values, though higher values are still present and carry meaningful information. The shape resembles an exponential distribution, suggesting that applying a natural logarithmic transformation would normalise the data to some extent, enabling easier modelling. Figure 2b shows the distribution of log-transformed values, $\ln S_a(T)$, clearly this is an improvement.

Since the response spectrum in an analytical sense is continuous, modelling it computationally requires discrete sampling over a range of period values, T . The proposed dataset contains a sufficient range of periods from $T = 0.0$ s, peak ground acceleration (PGA), to $T = 20.0$ s, 114 total values with a higher concentration towards lower periods. For this project, the dataset response spectrum was resampled to 91 periods up to $T = 5.0$ s, increasing the sample rate between $T = 0.5$ s and $T = 1.5$ s via linear interpolation.

Beyond the response spectrum, other features of the dataset are also used by the proposed model, particularly for latent space regularisation at both event-level and sample-level. Earthquake magnitude M and hypocentre depth Z_{hyp} were used for event-level regularisation, excluding events where $Z_{\text{hyp}} \geq 21$ km. Rupture distance R_{rup} and near-surface shear-wave velocity V_{s30} were used for sample-level regularisation, excluding samples where $R_{\text{rup}} \geq 300$ km or $V_{s30} \geq 750$ ms^{-1} . The filters were applied to remove anomalies, also aiding visualisation of results, and reduced the total number of samples to 18,493 with 573 distinct events.

Next, consider the distribution of samples with respect to the regularisation features listed above. For instance, while lower-magnitude earthquakes are more common in the dataset, the overall distribution of earthquake magnitudes appears bimodal (Fig. 3a). This could pose a challenge for the model when learning from medium-magnitude events (around $M = 5.0$), as their embeddings may be drawn toward either of the two dominant modes. This represents a form of multivariate continuous class imbalance, as it involves imbalanced distributions across multiple continuous variables. To address this, a technique known as *kernel density estimate (KDE) undersampling* was employed [40]. This involves fitting a multivariate KDE to the feature space and selectively drawing samples from a weighted uniform distribution, where weights are inversely proportional to the estimated density. Figure 3 shows the effect of this undersampling technique.

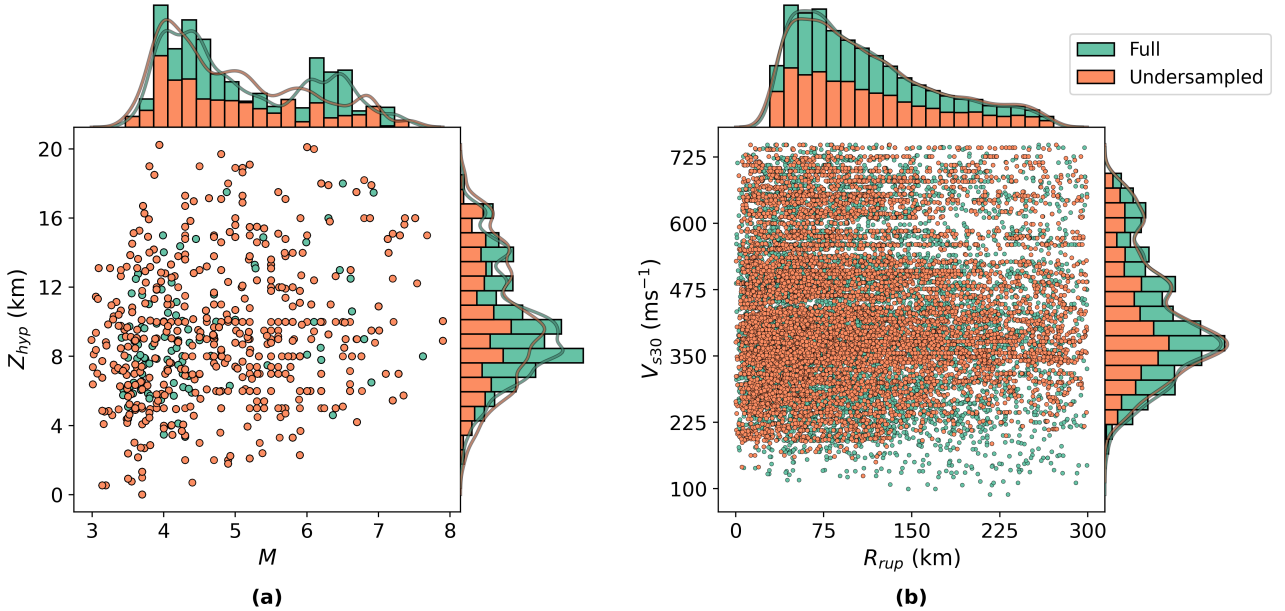


Figure 3: The distribution of samples in the dataset by (a) M and Z_{hyp} , and (b) R_{rup} and V_{s30} , overlaid with an 50% resampled subset using KDE undersampling.

An 80% KDE undersampling was done to the dataset following filtering. This reduced the total number of samples to 14,908 with 553 distinct events.

3.2 Base Framework Implementation

From previous research, it is clear that a VAE approach would be most appropriate, initially building upon previous work by Fayaz et al. (2025) [6]. The VAE framework utilises two neural networks, an encoder and a decoder, linked via an intermediate latent space capturing a low-dimensional representation of the data. The weights of the two networks are traditionally trained by maximising the evidence lower bound (ELBO), a tractable lower bound on the log-likelihood of the observed data under the generative model. This approach considers the marginal probability of this likelihood over the latent embeddings, balancing reconstruction accuracy with regularisation of the latent distribution.

$$\log p_\theta(\mathbf{x}) \geq \mathbb{E}_{q_\phi(\mathbf{z} | \mathbf{x})} [\log p_\theta(\mathbf{x} | \mathbf{z})] - \text{KL}(q_\phi(\mathbf{z} | \mathbf{x}) \parallel p(\mathbf{z})) \quad (1)$$

Here, $p()$ is the prior distribution, $q_\phi()$ is the approximate posterior distribution, $p_\theta()$ is the likelihood, θ are the parameters of the decoder, ϕ are the parameters of the encoder, $\mathbf{x}$ is some observed data, and $\mathbf{z}$ is the latent embedding of $\mathbf{x}$.

A reparameterisation trick is used to backpropagate through stochastic sampling by having the encoder output the parameters of a normal distribution and sampling from this to produce embeddings. Note that the outputs of the encoder are typically $\boldsymbol{\mu}_\phi(\mathbf{x})$ and $\ln \boldsymbol{\sigma}_\phi(\mathbf{x})^2$, so the output shape is $2d$ for d latent dimensions.

$$\mathbf{z} = \boldsymbol{\mu}_\phi(\mathbf{x}) + \boldsymbol{\epsilon} \cdot \boldsymbol{\sigma}_\phi(\mathbf{x}), \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \quad (2)$$

This approach accounts for random variation in the regenerated samples from the latent space. Furthermore, Eq. 2 allows us to rewrite the second term of Eq. 1 as a closed form expression, assuming the prior distribution is a standard Gaussian (typical of VAEs):

$$\begin{aligned} \text{KL}(q_\phi(\mathbf{z} | \mathbf{x}) \parallel p(\mathbf{z})) &= \text{KL}(\mathcal{N}(\boldsymbol{\mu}_\phi(\mathbf{x}), \boldsymbol{\sigma}_\phi^2(\mathbf{x})) \parallel \mathcal{N}(\mathbf{0}, \mathbf{I})) \\ &= \frac{1}{2} \sum_{i=1}^d \left[\log \left(\frac{1}{\boldsymbol{\sigma}_\phi^2(\mathbf{x})_i} \right) + \frac{\boldsymbol{\sigma}_\phi^2(\mathbf{x})_i}{1} + \frac{(\boldsymbol{\mu}_\phi(\mathbf{x})_i - 0)^2}{1} - 1 \right] \\ \text{KL}(q_\phi(\mathbf{z} | \mathbf{x}) \parallel p(\mathbf{z})) &= \frac{1}{2} \sum_{i=1}^d \left[\boldsymbol{\sigma}_\phi^2(\mathbf{x})_i + (\boldsymbol{\mu}_\phi(\mathbf{x})_i)^2 - \ln \boldsymbol{\sigma}_\phi^2(\mathbf{x})_i - 1 \right] \end{aligned} \quad (3)$$

Finally, since the first term of Eq. 1 is concerned with reconstruction, we can approximate it using the mean squared error function (MSE). By considering the average of these terms over all samples in batch B gives us the following expression for the loss function of the VAE:

$$\mathcal{L}(\theta, \phi; B) = \underbrace{\frac{1}{|B|} \sum_{\mathbf{x} \in B} \|\mathbf{x} - \hat{\mathbf{x}}\|^2}_{L_{recon}} + \underbrace{\frac{1}{|B|} \sum_{\mathbf{x} \in B} \left(\frac{1}{2} \sum_{i=1}^d \left[\boldsymbol{\sigma}_\phi^2(\mathbf{x})_i + (\boldsymbol{\mu}_\phi(\mathbf{x})_i)^2 - \ln \boldsymbol{\sigma}_\phi^2(\mathbf{x})_i - 1 \right] \right)}_{L_{KL}} \quad (4)$$

where $\hat{\mathbf{x}}$ is the reconstructed sample, output of the decoder.

To take this model one step further and bring it in line with the recent work by Fayaz et al (2025) [6], we can add the following regularisation loss term to encourage meaningful latent variables with respect to other features in the dataset:

$$L_{reg} = \frac{1}{|B|^2} \sum_{(\mathbf{x}, \mathbf{y}) \in B \times B} \exp \left(- \sum_{f \in F} \frac{\|f_{\mathbf{x}} - f_{\mathbf{y}}\|^2}{\tau_f} \right) \cdot \|\mathbf{z}_{\mathbf{x}} - \mathbf{z}_{\mathbf{y}}\|^2 \quad (5)$$

where $\mathbf{x}, \mathbf{y}$ are pairwise samples from the batch, F is the set of regularisation features, $f_{\mathbf{x}}$ is the value of feature f for sample $\mathbf{x}$, τ_f is the standard deviation across all samples for feature f , and $\mathbf{z}_{\mathbf{x}}$ is the embedding of sample $\mathbf{x}$. The addition of this term is used to ensure *similar samples are embedded closely* in the latent space. Combining equations 4 and 5, and adding corresponding weights to each term to control their contribution, gives the final expression for the batch-wise loss of the base VAE framework:

$$\mathcal{L}(\phi, \theta; B, F, W_{recon}, W_{KL}, W_{reg}) = W_{recon} L_{recon} + W_{KL} L_{KL} + W_{reg} L_{reg} \quad (6)$$

3.3 Latent Space Adaptations

3.3.1 Hierarchical Embeddings

This project considers correlations between samples belonging to the same earthquake event by giving each group, earthquake event, a distinct prior distribution. A further consideration can then be taken to ensure that the distribution of these priors is maintained. More specifically, the mean of these priors comes from a standard normal whilst the variance comes from a half-normal. The effect of this is a hierarchical prior distribution where for some sample embedding z belonging to the group indexed at g we have:

$$\mathbf{z} \sim \mathcal{N}(\boldsymbol{\mu}_g, \boldsymbol{\sigma}_g^2), \quad \text{s.t.} \quad \boldsymbol{\mu}_g \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad \boldsymbol{\sigma}_g^2 \sim \text{half-}\mathcal{N}(\mathbf{0}, \mathbf{I})$$

The visual intuition for this is that, in the broader picture, all embeddings follow a standard normal distribution, but when focussing upon a particular event, all embedded samples are clustered around some smaller normal distribution. This consequently encodes spatial correlation within the latent space; one can sample a point to represent an event and then sample the closely surrounding space to generate response spectra which should reflect multiple recordings of this event from a range of different sites.

To implement the hierarchical prior, the original KL-divergence loss term, L_{KL} , was adjusted and a further KL-divergence for the hyper-prior was added. To keep the model somewhat simpler, the group-level variance, $\boldsymbol{\sigma}_g = \sigma_g$, was fixed (scalar constant for all dimensions) which means that this variable no longer comes from a half-normal and it did not need to be regularised in the loss. Hence, the two KL-divergence loss terms are:

$$L_{KL} = \frac{1}{|B|} \sum_{g \in G} \sum_{\mathbf{x} \in G_g} \left(\frac{1}{2} \sum_{i=1}^d \left[\ln \left(\frac{\sigma_g^2}{\sigma_\phi^2(\mathbf{x})_i} \right) + \frac{\sigma_\phi^2(\mathbf{x})_i}{\sigma_g^2} + \frac{(\boldsymbol{\mu}_\phi(\mathbf{x})_i - \boldsymbol{\mu}_{g,i})^2}{\sigma_g^2} - 1 \right] \right) \quad (7)$$

$$\text{and } L_{KL, hyp} = \frac{1}{2} \sum_{i=1}^d [\sigma_{G,i}^2 + \boldsymbol{\mu}_{G,i}^2 - \ln \sigma_{G,i}^2 - 1] \quad (8)$$

where G is the set of group indices, $\boldsymbol{\mu}_G$ and $\boldsymbol{\sigma}_G^2$ are the empirical mean and variance of $\{\boldsymbol{\mu}_g : g \in G\}$, and G_g is the set of samples belonging to the group indexed at g .

Since the objective now groups samples by event, global regularisation by sample-level features is not valid since this would inherently pull groupings apart. Instead, one can implement two levels of regularisation, given the current two tier hierarchy. Globally, regularisation can be applied to the group means using event-level features such as earthquake magnitude. Within each group, sample-level features such as rupture distance can be used to regularise at the lower level. These two forms of regularisation are analogous to the hyper-prior and prior distributions discussed above.

For the group-level regularisation, a similar subset of features, F_{hyp} , can be defined, and differences between group means should be used in place of the differences between sample embeddings from Eq. 5. It is important to consider what exactly this term now encourages; it ensures similar group means are pulled close to one another. However, making this adjustment has added implications. In the current set up, there is little to stop the model from overlapping group means with identical event-level features, there is nothing to encourage spread. The intuitive solution to this problem is to add another regularisation term which *encourages group means of different events to be far apart*. More formally, we can define a hyper-regularisation loss term as:

$$L_{reg, hyp} = \frac{1}{|G|^2} \sum_{(g,h) \in G \times G} \left[\underbrace{e^{-F_{g,h}} \cdot \|\boldsymbol{\mu}_g - \boldsymbol{\mu}_h\|^2}_{\text{Cohesion}} + \underbrace{e^{(-\|\boldsymbol{\mu}_g - \boldsymbol{\mu}_h\|^2)} \cdot (F_{g,h} + s)}_{\text{Dispersion}} \right] \quad (9)$$

$$\text{where, } F_{g,h} = \sum_{f \in F_{hyp}} \frac{\|f_g - f_h\|^2}{\tau_f}$$

The two parts to Eq. 9 are labelled as the *cohesion* and *dispersion* parts since the former encourages group means to be close whilst the latter encourages them to spread. An additional parameter s is added here to ensure a minimum spread between all group means, discouraging overlap between events with identical features.

Similarly, since sample-level regularisation involves only as many samples as there are per group, a dispersion term could also be useful here. Note that a base dispersion level is not required since overlapping samples is not a major concern.

$$L_{reg} = \frac{1}{|B|^2} \sum_{g \in G} \sum_{(\mathbf{x}, \mathbf{y}) \in G_g \times G_g} \left[e^{\left(-\sum_{f \in F} \frac{\|f_{\mathbf{x}} - f_{\mathbf{y}}\|^2}{\tau_f}\right)} \cdot \|\mathbf{z}_{\mathbf{x}} - \mathbf{z}_{\mathbf{y}}\|^2 + e^{(-\|\mathbf{z}_{\mathbf{x}} - \mathbf{z}_{\mathbf{y}}\|^2)} \cdot \sum_{f \in F} \frac{\|f_{\mathbf{x}} - f_{\mathbf{y}}\|^2}{\tau_f} \right] \quad (10)$$

where only differences **within** each group are considered.

Since there are a moderate number of loss terms, the total batch-wise loss is better expressed through a vector product:

$$\mathcal{L}(\phi, \theta, \{\boldsymbol{\mu}_g : g \in G\} ; B, F, F_{hyp}, \mathbf{W}, G) = \mathbf{W}^T \mathbf{L} \quad (11)$$

$$\text{where } \mathbf{W} = \begin{bmatrix} W_{recon} \\ W_{KL} \\ W_{reg} \\ W_{KL, hyp} \\ W_{reg, hyp} \end{bmatrix} \quad \text{and} \quad \mathbf{L} = \begin{bmatrix} L_{recon} \\ L_{KL} \\ L_{reg} \\ L_{KL, hyp} \\ L_{reg, hyp} \end{bmatrix}$$

3.3.2 Hyperbolic Geometry

Given the complexity of the proposed latent space, embedding nearly 15,000 samples separated into 553 groups regularised at two levels, it may be challenging for standard Euclidean space to capture the necessary variance required to represent this. Hyperbolic geometry, in the form of the Poincaré ball, incorporates negative intrinsic curvature to fit infinite Euclidean space into an open ball of radius $1/\sqrt{c}$, where $-c$ is the curvature. Utilising hyperbolic geometry involves redefining elementary mathematical concepts such as the distance metric, addition, and exponential and logarithmic mappings. The formulation results in a space where distances grow exponentially as one approaches the edge of the ball, hence the ball is open. It has been shown that this approach yields better latent structure when applied to hierarchically related data [37, 38].

More formally, the Poincaré ball space of negative curvature c is defined as:

$$\mathbb{B}_c^d = \left\{ \mathbf{x} \in \mathbb{R}^d : \|\mathbf{x}\| < \frac{1}{\sqrt{c}} \right\} \quad (12)$$

equipped with the **distance metric**:

$$d_p^c(\mathbf{u}, \mathbf{v}) = \frac{1}{\sqrt{c}} \cosh^{-1} \left(1 + 2c \frac{\|\mathbf{u} - \mathbf{v}\|^2}{(1 - c\|\mathbf{u}\|^2)(1 - c\|\mathbf{v}\|^2)} \right) \quad (13)$$

where $\|\mathbf{x}\|$ denotes the standard norm of a vector, and $\mathbf{u}, \mathbf{v} \in \mathbb{B}_c^d$ [38, 41].

Since the notion of distance is different in hyperbolic space, other concepts such as **addition**

change accordingly. The following is called *Möbius addition*:

$$\mathbf{u} \oplus_c \mathbf{v} = \frac{(1 + 2c\langle \mathbf{u}, \mathbf{v} \rangle + c\|\mathbf{v}\|^2)\mathbf{u} + (1 - c\|\mathbf{u}\|^2)\mathbf{v}}{1 + 2c\langle \mathbf{u}, \mathbf{v} \rangle + c^2\|\mathbf{u}\|^2\|\mathbf{v}\|^2} \quad (14)$$

where $\langle \mathbf{x}, \mathbf{y} \rangle$ denotes the standard dot product operation between two vectors [38, 41].

Building upon this framework, one can derive expressions for the **exponential** and **logarithm** maps within this space:

$$\exp_{\mathbf{x}}^c(\mathbf{u}) = \mathbf{x} \oplus_c \left(\tanh \left(\sqrt{c} \frac{\lambda_{\mathbf{x}}^c \|\mathbf{u}\|}{2} \right) \frac{\mathbf{u}}{\sqrt{c}\|\mathbf{u}\|} \right) \quad (15)$$

$$\log_{\mathbf{x}}^c(\mathbf{u}) = \frac{2}{\sqrt{c}\lambda_{\mathbf{x}}^c} \tanh^{-1} \left(\sqrt{c} \|\mathbf{x} \oplus_c \mathbf{u}\| \right) \frac{-\mathbf{x} \oplus_c \mathbf{u}}{\|\mathbf{x} \oplus_c \mathbf{u}\|} \quad (16)$$

where $\lambda_{\mathbf{x}}^c = \frac{2}{1 - c\|\mathbf{x}\|^2}$ is the *conformal factor* at point $\mathbf{x}$ [38, 41].

Following these mathematical definitions, one next needs to derive some notion of a **normal distribution** in the Poincaré ball model. There are a few different approaches that one might take, the *Riemannian normal* and the *wrapped normal* being the most intuitive two [38]. This project shall focus on the latter due to its simplicity. The wrapped normal can be defined using a normal distribution in Euclidean space and mapping it to $\mathbb{B}_c^d$ via the exponential map defined in Eq. 15 [38, 42]. The underlying theory behind this approach is that the Euclidean space used for the original sampling can be treated as the tangent plane to $\mathbb{B}_c^d$ at the point $\boldsymbol{\mu}$, $T_{\boldsymbol{\mu}}\mathbb{B}_c^d$, since any tangent plane in hyperbolic space is isomorphic to $\mathbb{R}^d$ [43]. The exponential map then projects this point onto $\mathbb{B}_c^d$. Samples from the distribution can hence be taken as:

$$\mathbf{z} = \exp_{\boldsymbol{\mu}}^c \left(\frac{\mathcal{N}(\mathbf{0}, \boldsymbol{\sigma})}{\lambda_{\boldsymbol{\mu}}^c} \right) \quad (17)$$

To begin transforming the current Euclidean latent space of the model to a Poincaré ball, there is one final definition required. Since the posterior and prior distributions will now be wrapped normals, an approximation of the KL-divergence between these is required. Nagano et al (2019) [42] took an approach by approximating $KL(q_{\phi}(\mathbf{z} | \mathbf{x}) \| p_g(\mathbf{z})) \approx \log q_{\phi}(\mathbf{z} | \mathbf{x}) - \log p_g(\mathbf{z})$, for each data point. This is indeed valid although since each data point is associated with a distinct posterior distribution and only one sample is taken from the latent space from each posterior, this could be considered quite a weak approximation. However, given the simplicity of the posteriors, often with significantly small variances, and the large number of data points, this concern is minor. Therefore, the remaining challenge is to compute or approximate the probability density of observing $\mathbf{z}$ under both the posterior and the prior distributions. To do this, we can use the probability density function of the normal distribution used to sample the point in the tangent plane, $\mathcal{N}(\log_{\boldsymbol{\mu}}^c(\mathbf{z}) \cdot \lambda_{\boldsymbol{\mu}}^c | \mathbf{0}, \boldsymbol{\sigma})$, and then scale it by the appropriate change of variables factor given in [38]. This gives us the expression:

$$\mathcal{WN}_c(\mathbf{z} | \boldsymbol{\mu}, \boldsymbol{\sigma}) = \mathcal{N}(\lambda_{\boldsymbol{\mu}}^c \log_{\boldsymbol{\mu}}^c(\mathbf{z}) | \mathbf{0}, \boldsymbol{\sigma}) \cdot \left(\frac{\sqrt{c} d_p^c(\boldsymbol{\mu}, \mathbf{z})}{\sinh(\sqrt{c} d_p^c(\boldsymbol{\mu}, \mathbf{z}))} \right)^{d-1} \quad (18)$$

Details on how the Poincaré ball KL-divergence loss term was incorporated are included in appendix A.

The architectures of the encoder and decoder networks shall predominantly feature fully-connected layers computing affine transformations in Euclidean space, these shall remain Euclidean for simplicity. Therefore, one final consideration is required when mapping between the Euclidean networks and the Poincaré ball latent space. For the encoder, we can keep the log variance parts of the output the same since these are only used in the tangent space sampling, and map the mean parts to the space via the exponential map at $\mathbf{0}$, as was done in [37].

The decoder instead requires some form of conversion layer at the start to convert from hyperbolic geometry to some Euclidean or scalar values. The simplest approach is to consider what the analogue

to affine transformations are in the Poincaré ball space. In the Euclidean case, an affine transformation can be generalised as $f_{\mathbf{a},\mathbf{p}}(\mathbf{z}) = \langle \mathbf{a}, \mathbf{z} - \mathbf{p} \rangle$ where $\mathbf{a}, \mathbf{p} \in \mathbb{R}^d$ are the *orientation* and *offset* parameters respectively. Ganea et al (2018) [41] introduced the following Poincaré ball analogue to this function:

$$f_{\mathbf{a},\mathbf{p}}^c(\mathbf{z}) = \text{sign}(\langle \mathbf{a}, \log_{\mathbf{p}}^c \mathbf{z} \rangle_{\mathbf{p}}) \|\mathbf{a}\|_{\mathbf{p}} d_{\mathbf{p}}^c(\mathbf{z}, H_{\mathbf{a},\mathbf{p}}^c)$$

The *gyroplane layer* captures this form of affine transformation, allowing for a mapping from hyperbolic space to Euclidean in the first layer of the decoder. The layer has trainable parameters in the form of u pairs of vectors $\mathbf{p}$ and $\mathbf{a}$, each pair outputting the following:

$$d_p^c(\mathbf{z}, H_{\mathbf{a},\mathbf{p}}^c) = \frac{1}{\sqrt{c}} \cdot \sinh^{-1} \left(\frac{2\sqrt{c}|\langle -\mathbf{p} \oplus_c \mathbf{z}, \mathbf{a} \rangle|}{(1 - c - \|\mathbf{p} \oplus_c \mathbf{z}\|^2)\|\mathbf{a}\|} \right) \quad (19)$$

In summary, the following changes were required to convert the Euclidean model to a Poincaré ball model:

- Change the re-parameterisation function to sample from $\mathcal{WN}(\exp_0 \boldsymbol{\mu}, \exp(\ln \boldsymbol{\sigma}^2))$, where $\boldsymbol{\mu}$ and $\ln \boldsymbol{\sigma}^2$ are the outputs of the encoder.
- Change the first layer of the decoder to a gyroplane layer.
- Update the KL-divergence loss term $L_{KL} = \log q_{\phi}(\mathbf{z} | \mathbf{x}) - \log p(\mathbf{z} | \mathbf{x})$, using equation 13 to compute the probabilities.
- Update the regularisation loss terms L_{reg} to use Eq. 13, $d_p^c(\cdot, \cdot)$ for computing the distance between group means and between embedded samples.

A decision was made to keep the group means stored in Euclidean space, mapping to the Poincaré ball via $\exp_0^c(\cdot)$ only when needed, since this ensures that the group means remain within the bounds of the open ball. Hence, no change was required in the hyper-prior KL-divergence loss term besides reducing the variance to ensure a balance of spread throughout the curved latent space since the curvature distorts how a standard normal distribution looks.

3.3.3 Group Mean Initialisation

Upon first implementing the hierarchical prior, there was no dispersion term in L_{reg} (Eq. 9), just the original adhesion term. Only regularising via adhesion proved to be challenging. Consider first of all the various approaches one might take to initialising the means of these group prior distributions. Here are two intuitive approaches explored:

1. Randomly sample each one from a standard normal distribution, since this is the target hyper-prior.
2. Initialise all as zero.

The issue with approach 1 is that a random initialisation leads to an approximately even distribution of regularisation features across the latent space. This means that all group means would naturally be pulled towards the origin since an even spread of similar group means exist and the majority of these would be relatively closer to or on the other side of the origin. However, the hyper-prior KL-divergence loss term would discourage this mass movement since it would reduce variance below the initialised and desired value. The resulting effect is that the group means simply do not move and no regularisation can be attained.

The issue with approach 2 is that L_{reg} in Eq. 5 is computed using the pairwise distances between all group means which, if zero, sets this entire term to zero. Moreover, this represents a local maximum of this loss term since movement in all directions corresponds to an equal drop in the loss. Hence,

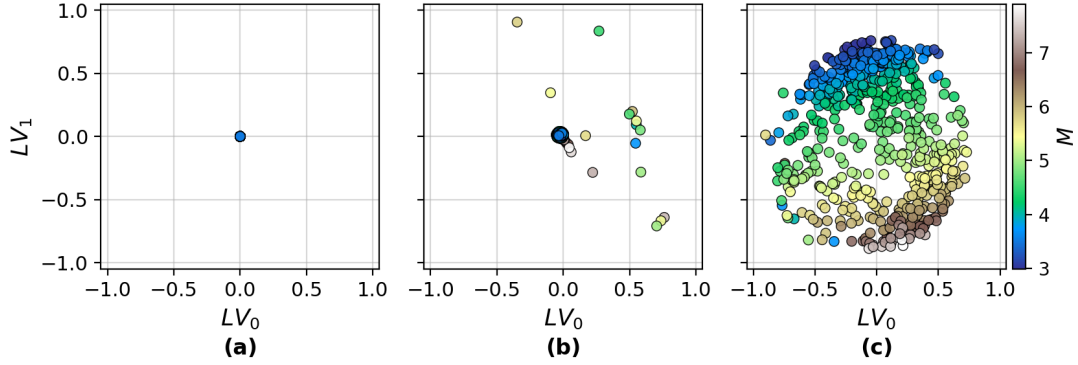


Figure 4: (a) The group means at epoch 0 (initialisation), they have been initialised using a normal distribution with variance 10^{-6} . (b) The group means after 100 epochs **without the dispersion term**, there is a high concentration of group means at the origin. (c) The group means after 100 epochs **with the dispersion term**. Points have been colour coded based on M .

in practice the group means are only initialised close to the origin to prompt some movement. The remaining problem is that the KL-divergence between the embedded samples and the group priors, at least in the early epochs, can be trivially solved by the model embedding all samples close to the origin too, this is not desirable. Furthermore, the hyper-prior KL-divergence only requires the variance of the distribution to be increased. The model could once again find a trivial solution to this by pushing a small number of these group means far away from the origin, artificially inflating the variance, this is also not desirable.

An approach to overcome this problem is to introduce some further loss term which penalises group means being clustered together. Given that the original L_{reg} term kept similar points close, it could easily be transformed to keep different points far apart. The resulting effect, shown in Figure 4, is a well structured prior for the embedded samples to follow.

Hence, group means can safely be initialised using approach 2.

3.4 Neural Network Architecture

The design of the encoder and decoder networks in a VAE requires careful consideration as the level of complexity can have a significant effect on the latent space structure as well as the reconstruction accuracy [44]. Given the complexity of the latent space proposed in this project, the encoder naturally needs to have a high degree of flexibility to embed samples in the required way. Also, it is common practice to approximately mirror the architectures of the encoder and decoder networks [44, 45], so both shall be reasonably deep architectures.

One approach that was considered for the architecture was one-dimensional convolutional layers since the response spectrum is a continuous feature measured at discrete intervals, similar to a digital image, and success has previously been seen applying CNNs to earthquake spectra [16, 23]. However, intricate detail between neighbouring periods is lost very quickly with this approach. Unlike images, where local smoothing or downsampling typically preserves the overall semantic content, response spectra are highly sensitive to local variations, especially between adjacent periods. These subtle fluctuations can correspond to critical structural resonances, meaning that distortions through convolution may significantly alter the physical interpretation. Therefore, convolutional layers were not used for this project in an attempt to preserve intricate detail.

The proposed neural networks were composed almost purely of fully-connected layers, arranged in residual blocks to allow for a deeper, more complex design [46]. Note that since a Poincaré ball latent space is used, the first layer of the decoder is always a Gyroplane layer. A function was built to generate the required network architecture based on proportions of the number of dimensions in the response spectrum, $\mathbf{P}$, and number of residual blocks, N . The following procedure in algorithm 1 was

used, note that $\mathbf{P}$ is expected to have first entry 1.0 and monotonically decrease to some number > 0.0 . This procedure results in a network comprised of N residual blocks each containing two fully-connected layers followed by a series of $|\mathbf{P}| - N$ fully-connected layers (note that N must be $< |\mathbf{P}|$).

Algorithm 1 Build Residual Network

```

1: procedure BUILDRESIDUALNETWORK(spectrum_dim, latent_dim, name,  $N$ ,  $P$ )
2:   if name = Encoder then
3:     inputs  $\leftarrow$  Input(shape = (spectrum_dim,))
4:      $x \leftarrow$  Dense(spectrum_dim  $\cdot$   $P[0]$ , ReLU)(inputs)
5:   else
6:     inputs  $\leftarrow$  Input(shape = (latent_dim,))
7:     Reverse  $P$ 
8:      $x \leftarrow$  GyroPlane(spectrum_dim  $\cdot$   $P[0]$ , ReLU)(inputs)
9:   end if
10:  for  $n = 0$  to  $\min(N, |\mathbf{P}| - 1) - 1$  do
11:    shortcut  $\leftarrow$   $x$ 
12:     $x \leftarrow$  Dense(spectrum_dim  $\times$   $P[n]$ )( $x$ )
13:     $x \leftarrow$  Dense(spectrum_dim  $\times$   $P[n]$ )( $x$ )
14:     $x \leftarrow$  Add([ $x$ , shortcut])
15:     $x \leftarrow$  Activation(ReLU)( $x$ )
16:     $x \leftarrow$  Dense(spectrum_dim  $\times$   $P[n + 1]$ , ReLU)( $x$ )
17:  end for
18:  for  $n = N + 1$  to  $|\mathbf{P}| - 1$  do
19:     $x \leftarrow$  Dense(spectrum_dim  $\times$   $P[n]$ , ReLU)( $x$ )
20:  end for
21:  if name = Encoder then
22:    outputs  $\leftarrow$  Dense( $2 \times$  latent_dim)( $x$ )
23:  else
24:    outputs  $\leftarrow$  Dense(spectrum_dim)( $x$ )
25:  end if
26:  return Model(inputs, outputs, name)
27: end procedure

```

4 Results

For the following results, an instance of the proposed model was trained using the configuration and data train-test split described in appendix B.

4.1 Reconstruction Accuracy

4.1.1 Metrics

The reconstruction accuracy of the proposed model was validated via two primary metrics: The coefficient of determination, R^2 , (Fig. 5a) and index of agreement (IoA) (Fig. 5b). The former can be used to validate reconstruction accuracy per period T by considering the residuals between original and reconstructed values across all samples. Whereas, the latter can be used to validate reconstruction accuracy per sample by treating response spectra as time-series data and analysing the difference from the mean observed value at every period T .

By analysing the IoA values across distributions of samples, the feature values associated to the most challenging response spectra to model can be identified. In particular, Figure 5a suggests that the model performs consistently well for lower M earthquakes ($M = 3.0$ to 4.0) with mean IoA close

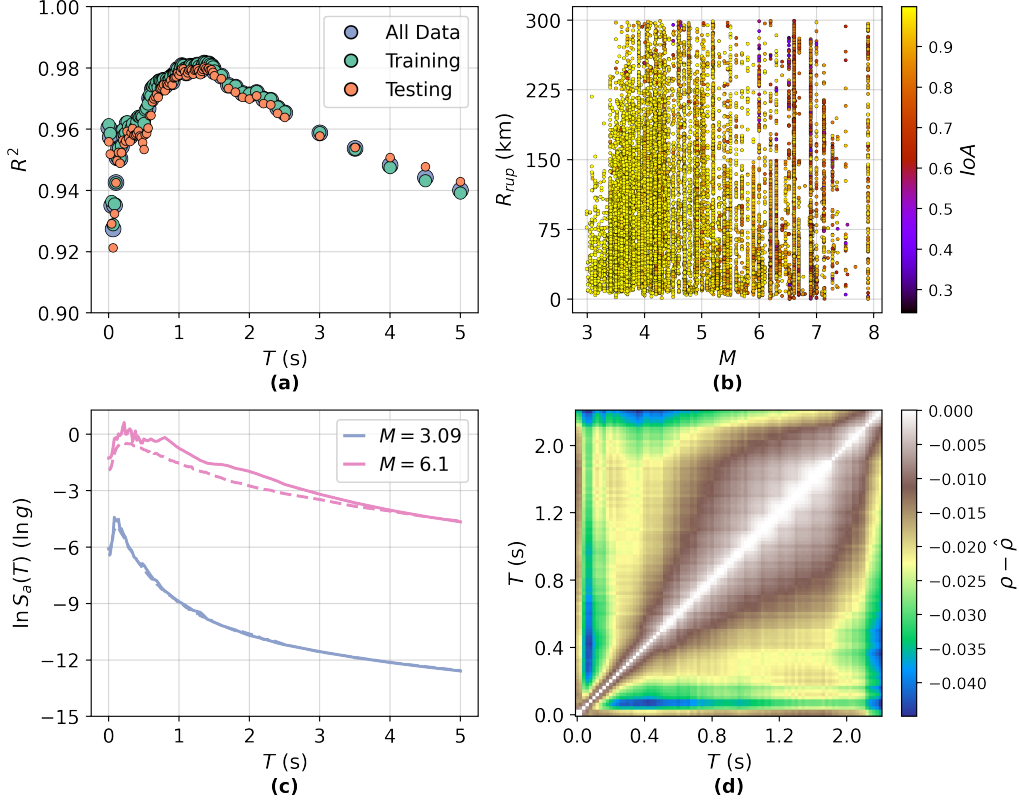


Figure 5: (a) A scatter plot of R^2 against the response spectrum period T . (b) A scatter plot of M against R_{rup} colour-coded by IoA. (c) Two example reconstructed response spectra (solid line is original and dashed line is reconstructed). (d) A heat map of the difference in cross-correlation coefficients for the true response spectra ρ and reconstructed response spectra $\hat{\rho}$ at each pair of spectral periods.

to 1.0, but struggles with higher M earthquakes ($M = 6.0$ to 8.0) with mean IoA around 0.75. The example reconstructed response spectra in Figure 5c also appears to support this. This performance difference likely reflects a combination of the physical complexity associated with high M earthquakes and the dataset imbalance despite KDE-undersampling. Further undersampling could have been done, however this would reduce the total number of samples, potentially harming the models performance for lower M earthquakes which are more common in practice.

The results in Figure 5a suggest strong global reconstruction with $R^2 > 0.92$ across all periods for the entire dataset with a mean value of 0.969. Reassuringly, R^2 values for the testing set are only slightly lower than the training set with a mean of 0.967 compared to 0.969, demonstrating good generalisability. It is important to note that most of the smallest R^2 values occur at very low periods which are naturally far more susceptible to noise from recordings and subsequently are more challenging to model [47]. Furthermore, most applications of response spectra are primarily interested in periods closer to 1.0 s since this approximately corresponds to the acceleration felt by a ten storey building [48], for which the model consistently achieves R^2 values at and above 0.97.

Given that response spectra represent periods of vibration within the recorded ground motion, there inherently exist correlations between acceleration at different periods. Preservation of these structural dependencies can be measured via the difference between the cross-correlation matrices computed across all samples in the dataset for true spectra and reconstructed spectra. Specifically, the Pearson correlation coefficient can be used between each pair of periods, a heat-map is used to visualise this difference in Figure 5d. The model effectively maintains cross-period correlations, particularly for mid-to-long periods ($T \in [0.5 \text{ s}, 2.0 \text{ s}]$), which are critical for seismology and engineering applications [49]. The maximum observed discrepancy occurs between $T = 0.06 \text{ s}$ and 5.0 s , with a correlation

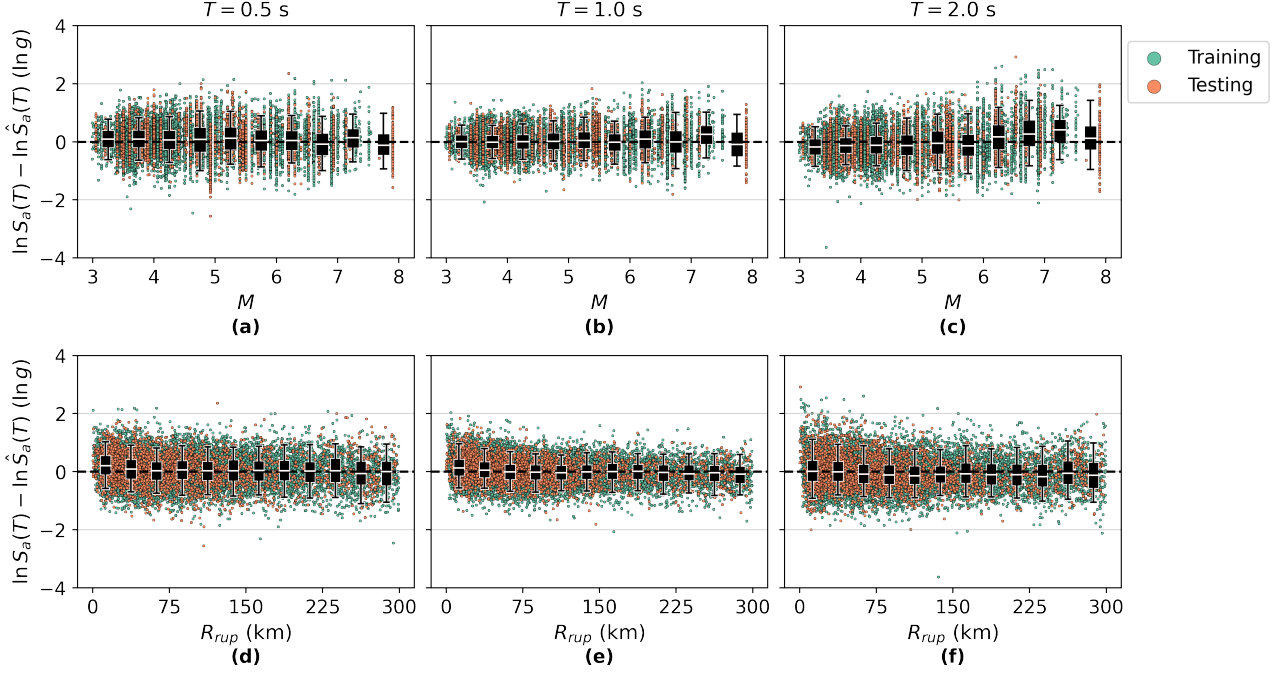


Figure 6: Scatter plots of the residual, $S_a(T) - \hat{S}_a(T)$ from each sample by M and R_{rup} for periods $T \in [0.5 \text{ s}, 1.0 \text{ s}, 2.0 \text{ s}]$.

difference of -0.0449 , indicating minimal error even at low period ranges.

4.1.2 Residual Analysis

Another way of analysing the reconstruction performance of the proposed model, and for identifying biases, is to plot the residuals (difference between the true response spectrum $S_a(T)$ and reconstructed response spectrum $\hat{S}_a(T)$) for all samples (Fig. 6).

It is reassuring to observe that residuals are predominantly symmetric about zero, suggesting that this error is likely due to random variation than poor modelling. At most, residuals tend to be $\in [-2, 2]$ with perhaps more variation in samples of lower R_{rup} , indicating that the model is not systematically over- or under-predicting across the dataset. There is a very minor asymmetry at $T = 2.0 \text{ s}$ for higher M samples where the predicted value is more commonly an underestimate. This could be due to how higher magnitude response spectra are generally higher across all periods compared to lower magnitude samples, paired with the preference of the model towards lower magnitude samples (Fig. 5a), perhaps a regression towards the mean. However, this is a very minor bias not present for all periods.

4.2 Latent Variable Exploration

4.2.1 Latent Dimensions

The number of latent dimensions d is an important aspect of any VAE model. In this project, all results were collected from a model using $d = 2$ due to improved interpretability of figures and little improvement to reconstruction performance for higher dimensions (Fig. 7). Another reason that d should be kept as small as possible is due to the Poincaré ball geometry. From Eq. 12 it is clear that for higher d there is an increased likelihood of samples being embedded close to the edge of the ball. This poses issues in the early stages of training which can occasionally lead to negative KL-divergence. More information on this issue can be found in appendix C.

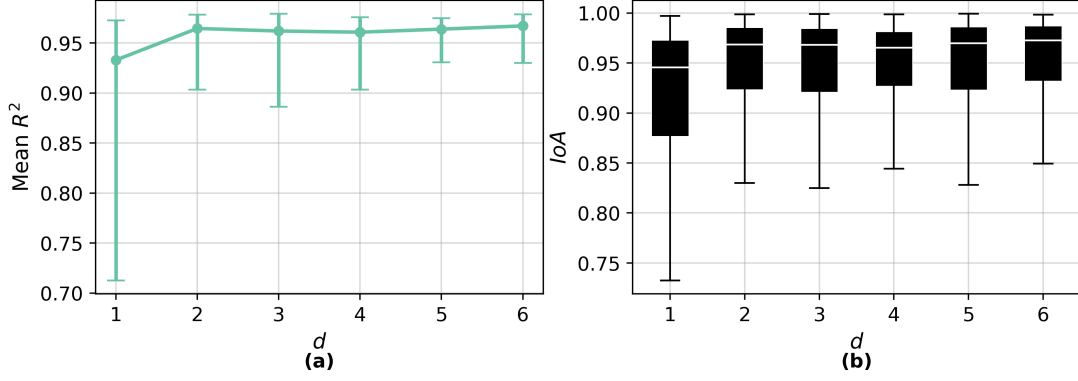


Figure 7: (a) Mean R^2 across all spectral periods against number of latent dimensions d . Error bars represent the minimum and maximum R^2 values across all spectral periods. (b) Distribution of IoA values across all samples against number of latent dimensions d , shown via box plots.

4.2.2 Feature Correlations

The proposed model incorporates both event-level features and sample-level features into the latent structure via regularisation loss terms (Eq. 9 and Eq. 10). The impact of these terms can be explored through visualisation of the latent space (Fig. 8) and computation of correlation coefficients (Tab. 1).

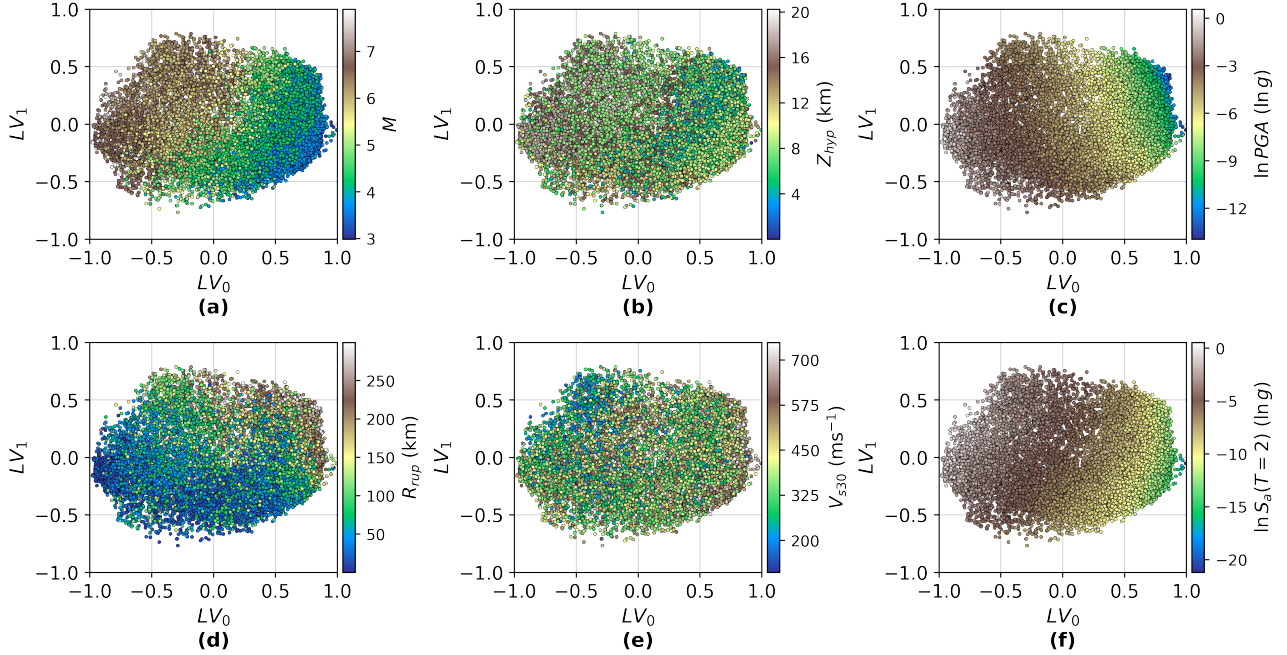


Figure 8: The latent space embeddings of all samples from both training and testing, colour-coded by (a) M , (b) Z_{hyp} , (c) log-transformed PGA, (d) R_{rup} , (e) V_{s30} , and (f) log-transformed $S_a(2)$.

It is highly reassuring to observe visual correlation in the latent space to some extent for all regularisation features, particularly prominent with M (Fig. 8a) and R_{rup} (Fig. 8d). Although some correlation is observed with Z_{hyp} (Fig. 8b) and V_{s30} (Fig. 8e), it is more ambiguous.

Pearson's correlation coefficient, ρ , is a common approach to quantifying linear correlations between features and latent variables, with values closer to 1.0 or -1.0 indicating stronger correlations. Given the hierarchical prior used in this project, it is also useful to compute ρ per group for all sample-level features and consider the mean. It is clear that M correlates most strongly with LV_0 in a negative direction with $\rho = -0.844$, this is consistent with Fig. 8a. There is also very strong correlation of

log-transformed PGA in this direction with $\rho = -0.932$, this is easily inferred from response spectra as it corresponds directly to $S_a(T = 0 \text{ s})$. It is interesting to observe how the R_{rup} correlation with LV_0 is the stronger of the two when analysed at group level with $\rho = 0.615$ but weaker when analysed globally. This could be due to how decreasing M results in a similar effect to increasing R_{rup} , hence at a local scale the model might intuitively align the R_{rup} correlation in the opposite direction to the M correlation. However, at a global scale this would not be as prevalent of a correlation due to "overlapping ends" of the group-level correlations, effectively cancelling each other out and reducing the overall metric.

Table 1: ρ computed between each feature, including log PGA , and both latent variables. Group-level refers to the mean ρ computed per group across all groups.

Feature	Global LV_0	Global LV_1	Group-level LV_0	Group-level LV_1
Magnitude M	-0.844	0.236	-	-
Hypocentre depth Z_{hyp}	-0.262	-0.091	-	-
Rupture distance R_{rup}	0.326	0.516	0.615	0.457
Shear-wave velocity V_{s30}	0.292	-0.050	0.240	-0.005
log PGA	-0.932	-0.290	-0.822	-0.627

An interesting observation present in all latent space embeddings generated by the model is the prominent spread along the axis parallel to the M correlation and lack of spread along the axis orthogonal to this. From Figure 3a, there doesn't appear to be any significant correlation between M and Z_{hyp} , so it is safe to assume that the hyper-regularisation used to arrange the group means would encourage correlation of Z_{hyp} orthogonally to M . This implies that the lack of spread is likely due to the model struggling to distinguish samples of high Z_{hyp} from samples of low Z_{hyp} , supported by previous research which found that M correlates much more strongly with response spectra than other features [18].

Given the hierarchical approach to regularisation, it is also useful to visualise feature correlations with latent variables for samples within groups. From Figure 9 it is clear that some correlation exists between samples within groups for both R_{rup} and V_{s30} . The distribution of samples in Fig. 9a for seem to support the stronger group-level correlation between R_{rup} and LV_0 seen in table 1. Also, the group-level correlation between V_{s30} and LV_0 is evident in the distributions of both sets of samples in Figure 9b.

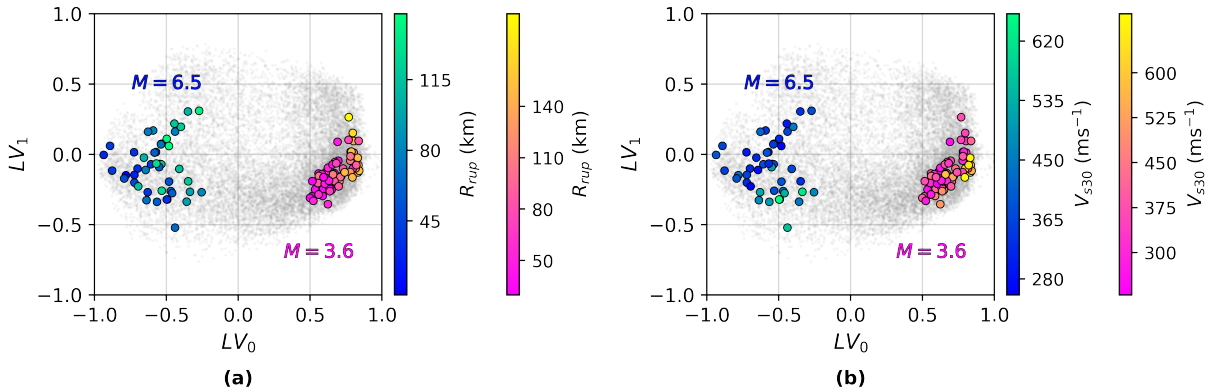


Figure 9: Latent embeddings from two events with magnitudes 3.6 and 6.5. Colour-coded by (a) R_{rup} , (b) V_{s30} .

4.3 Synthetic Data Generation

To make full use of the proposed framework, estimated feature values should be considered an output of the model alongside synthetic response spectra, so a clear approach to inference is required. The chosen approach was designed to best utilise the hierarchical latent structure by following the two stage sampling process (Fig. 1). This model uses pairs of points in the latent space, a sample $\mathbf{z}$ and the corresponding event centroid $\mathbf{C}$, within the following two neural network (NN) pipeline:

$$\underbrace{\mathbf{z}, \mathbf{C}}_{\text{Input data}} ; \underbrace{\mathbf{C} \rightarrow M, Z_{\text{hyp}}}_{\text{Event NN}} ; \underbrace{\mathbf{z}, M, Z_{\text{hyp}} \rightarrow R_{\text{rup}}, V_{s30}}_{\text{Sample NN}} ; \underbrace{M, Z_{\text{hyp}}, R_{\text{rup}}, V_{s30}}_{\text{Predicted features}}$$

Figure 10 shows the predicted feature values for all samples in the latent space using this approach. By visually comparing to Fig. 8 there is moderately strong reconstruction, particularly with respect to M for which an R^2 of 0.88 was achieved. The other features are predicted less consistently with R^2 values ≤ 0.31 , although similar correlations clearly exist.

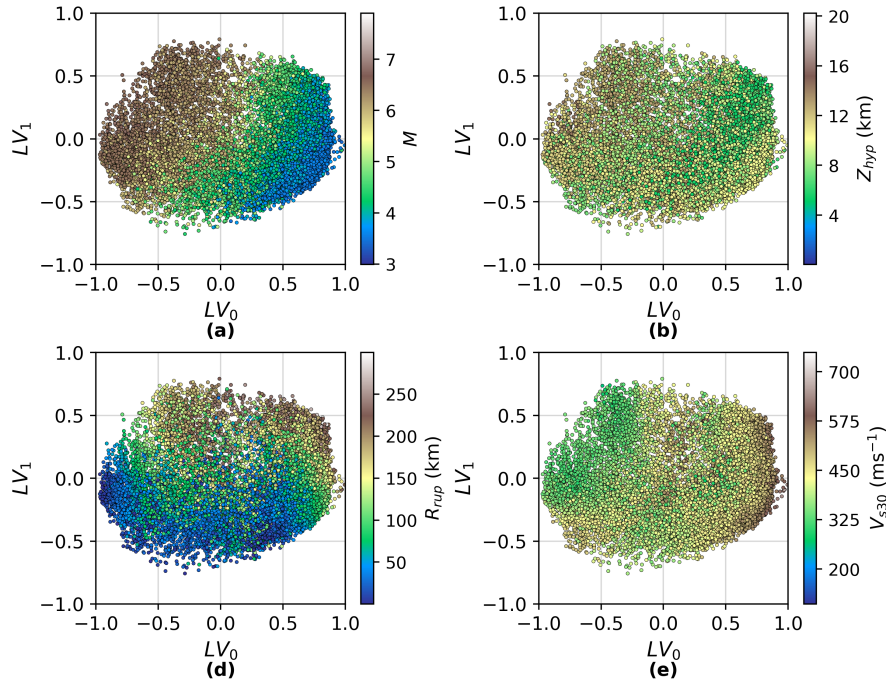


Figure 10: Latent space embeddings colour-coded by values of (a) M ($R^2 = 0.88$), (b) Z_{hyp} ($R^2 = 0.11$), (c) R_{rup} ($R^2 = 0.31$), (d) V_{s30} ($R^2 = 0.13$) predicted using the two-stage-NN approach.

The approach yields reasonable results considering that the correlations between these features and the latent variables are in themselves not hugely clear. By pairing this two-stage-NN with the trained latent space and decoder of the hierarchical VAE, some synthetic response spectra along with estimated feature values can be generated. Figure 11 demonstrates this for three samples of two distinct events. In both cases, the response spectra clearly demonstrate that increasing R_{rup} decreases $S_a(T)$ across the majority of periods, and by comparing the two events, it is also clear that decreasing M results in a similar effect.

4.4 Earthquake Early Warning Application

The proposed model enables seamless integration within a data-driven EEWS, serving as a generative decoder that reconstructs the full response spectrum from low-dimensional latent representations. Similar to Fayaz et al. (2025) [6], an on-site EEWS can be designed via an additional FFNN to

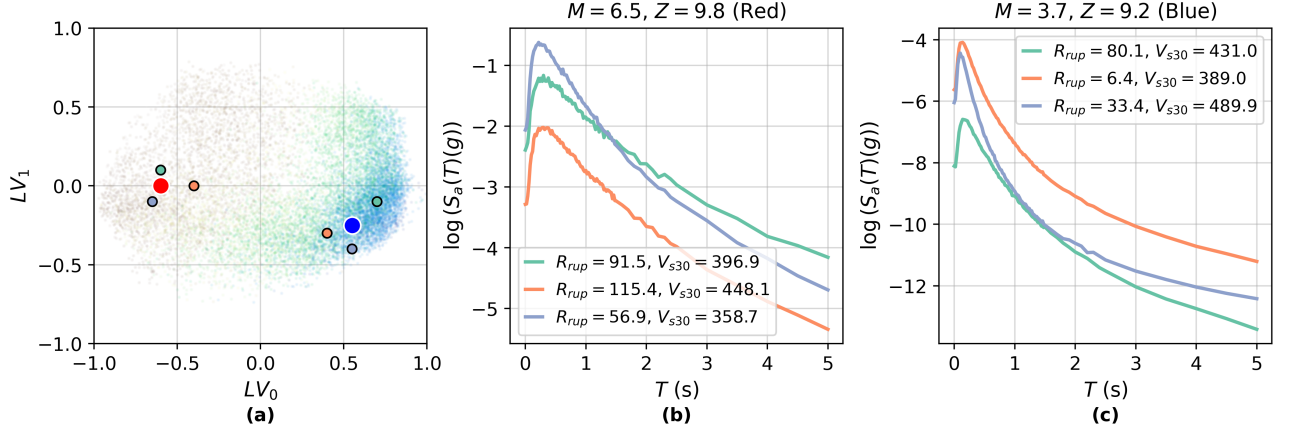


Figure 11: Examples of synthetic response spectra. (b) Shows spectra generated from the points surrounding the red centroid in (a). (c) Shows spectra generated from the points surrounding the blue centroid in (a).

predict a point in the latent space of the proposed VAE model, enabling real-time estimation of response spectra.

Figure 12a illustrates the full EEWS pipeline. An on-site seismic sensor monitors ground motion. Upon P-wave detection, a three-second processing window of recorded acceleration is used to compute the following set of IMs: Arias intensity (I_a), cumulative absolute velocity (CAV), peak ground velocity (PGV), PGA , peak ground displacement (PGD), mean response spectrum period (T_m), and significant duration (D_{595}). These are subsequently fed to an FFNN, predicting a point in the latent space which is decoded to an acceleration response spectrum and used to determine whether a warning should be issued. This approach effectively yields a full estimate of $S_a(T)$ within milliseconds of P-wave detection, without requiring information about source parameters or source-to-site geometry.

Training of the FFNN involved an 80:20 train-test split, with model performance evaluated through R^2 values comparing predicted and true latent variables. An additional PEER dataset was used for the inputs, mapping to samples in the original dataset via an ID field (*Record Sequence Number*) [39]. As shown in Figures 12b and c, the predicted LV_0 values show a strong linear trend with the true LV_0 , achieving $R^2 = 0.951$ across the full dataset. The predictive performance of LV_1 is less strong with $R^2 = 0.674$, this is likely due to the prominent correlation of LV_0 with respect to PGA . The training set achieved higher R^2 values than the testing set, average of 0.843 and 0.696 respectively, indicating possibly minor overfitting. However, this analysis purely serves to explore the model’s potential, and with further refinement and access to additional data, a more robust FFNN model could be developed.

5 Conclusion

This project proposes a hierarchical VAE for generating synthetic earthquake response spectra from a highly interpretable latent space from which both event-level and sample-level features can be inferred. The model achieves promising reconstruction results with a mean R^2 of 0.969 and symmetric residuals indicating that reconstruction error is likely due to random variation. The implementation of a Poincaré ball latent space was highly effective, with embeddings sufficiently covering the space and clear feature regularisation yielding ρ values of -0.844 for M with respect to LV_0 and 0.615 for R_{rup} with respect to LV_0 at group-level. The model was trained and tested on a significant portion of the NGA-West2 ground motion dataset and clearly generalised well.

A further application of the proposed model within an EEWS was also explored via the brief development of an FFNN. The approach effectively predicted points in the latent space from IMs computed using the first three seconds of ground motion, achieving R^2 of 0.951 and 0.674 for the two

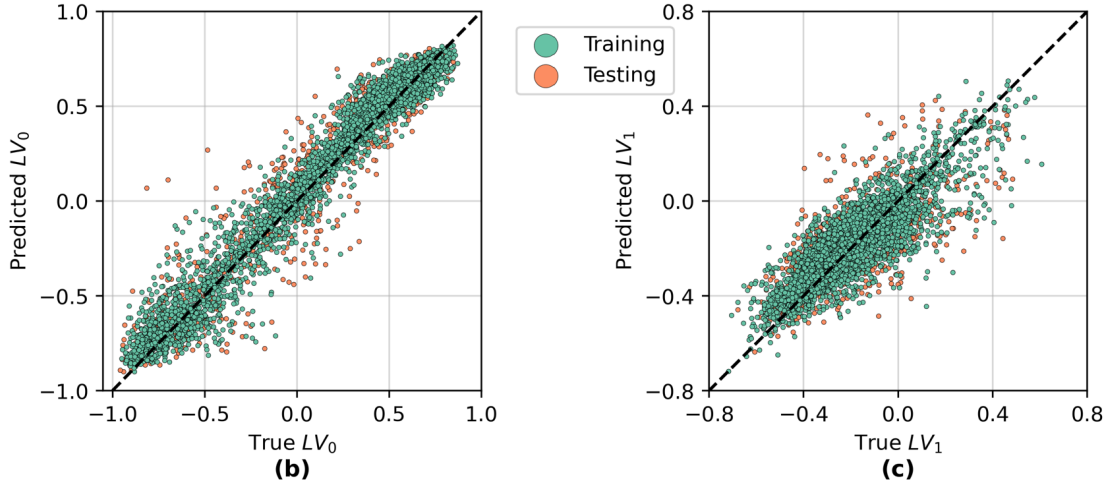
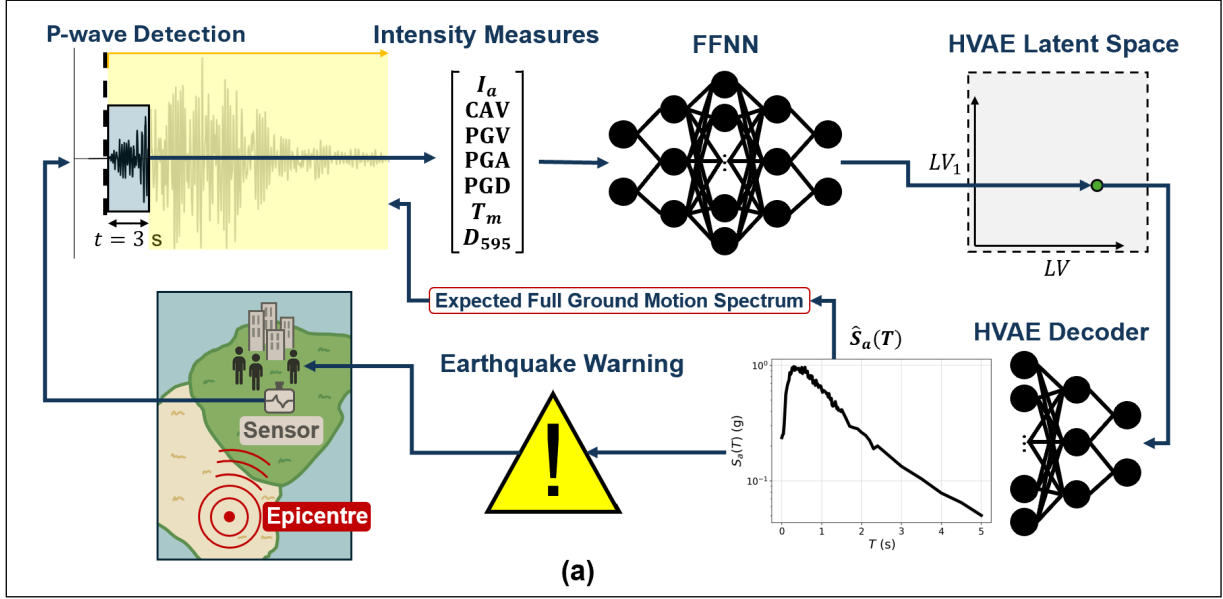


Figure 12: (a) Earthquake early warning pipeline: an on-site sensor monitors for any live ground motion. After P-wave detection, a three-second window of ground motion is used to compute a vector of IMs. The IM vector is fed to a pre-trained FFNN to estimate the corresponding point in the proposed VAE latent space. The estimated point is then decoded into the full response spectrum and used to conduct the final early warning. (b)-(c) True latent variables plotted against predicted latent variables from the FFNN, (b) z_0 ($R^2 = 0.951$ for all data) and (c) z_1 ($R^2 = 0.674$ for all data).

latent variables respectively. Hence, from a domain perspective, the chosen modelling approach has clear value in real-world applications.

Although the model is successful, further work could be done to investigate the use of other dataset features for regularisation, the introduction of further tiers in the hierarchy, and allowing for variable prior distribution variance σ_g^2 . This project highlights the positive impact of incorporating more physically meaningful attributes of the data into ML approaches within the domain of seismology.

6 Acknowledgments

I would like to thank my academic supervisor Dr Jawad Fayaz for his guidance and support throughout this project, and for sourcing both datasets used. I would also like to thank The University of Exeter, Department of Computer Science for the opportunity to undertake this project.

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Appendix A

For clarity on how the theory was translated into practice, some notes have been included here to discuss the considerations that were taken when implementing the wrapped normal probability density function in Python with TensorFlow. To reiterate, the probability density function of the wrapped normal distribution is:

$$\mathcal{WN}_c(\mathbf{z} \mid \boldsymbol{\mu}, \boldsymbol{\sigma}) = \mathcal{N}(\lambda_{\boldsymbol{\mu}}^c \log_{\boldsymbol{\mu}}^c(\mathbf{z}) \mid \mathbf{0}, \boldsymbol{\sigma}) \cdot \left(\frac{\sqrt{c} d_p^c(\boldsymbol{\mu}, \mathbf{z})}{\sinh(\sqrt{c} d_p^c(\boldsymbol{\mu}, \mathbf{z}))} \right)^{d-1}$$

Since this is only used to find the log probability, we only need to implement the following:

$$\begin{aligned} \log(\mathcal{WN}_c(\mathbf{z} \mid \boldsymbol{\mu}, \boldsymbol{\sigma})) &= \log(\mathcal{N}(\lambda_{\boldsymbol{\mu}}^c \log_{\boldsymbol{\mu}}^c(\mathbf{z}) \mid \mathbf{0}, \boldsymbol{\sigma})) + \log \left(\left(\frac{\sqrt{c} d_p^c(\boldsymbol{\mu}, \mathbf{z})}{\sinh(\sqrt{c} d_p^c(\boldsymbol{\mu}, \mathbf{z}))} \right)^{d-1} \right) \\ &= \log \left(\frac{1}{(2\pi)^{\frac{d}{2}} \boldsymbol{\sigma}^{\frac{1}{2}}} \exp \left(-\frac{1}{2} (\lambda_{\boldsymbol{\mu}}^c \log_{\boldsymbol{\mu}}^c(\mathbf{z}) - \mathbf{0})^T \boldsymbol{\sigma}^{-1} (\lambda_{\boldsymbol{\mu}}^c \log_{\boldsymbol{\mu}}^c(\mathbf{z}) - \mathbf{0}) \right) \right) + \log \left(\left(\frac{\sqrt{c} d_p^c(\boldsymbol{\mu}, \mathbf{z})}{\sinh(\sqrt{c} d_p^c(\boldsymbol{\mu}, \mathbf{z}))} \right)^{d-1} \right) \\ &= -\log \left((2\pi)^{\frac{d}{2}} \boldsymbol{\sigma}^{\frac{1}{2}} \right) - \frac{1}{2} (\lambda_{\boldsymbol{\mu}}^c \log_{\boldsymbol{\mu}}^c(\mathbf{z}))^T \boldsymbol{\sigma}^{-1} (\lambda_{\boldsymbol{\mu}}^c \log_{\boldsymbol{\mu}}^c(\mathbf{z})) + (d-1) \log \left(\frac{\sqrt{c} d_p^c(\boldsymbol{\mu}, \mathbf{z})}{\sinh(\sqrt{c} d_p^c(\boldsymbol{\mu}, \mathbf{z}))} \right) \end{aligned}$$

Since there are no covariances between the dimensions, this can be simplified to:

$$\begin{aligned} &= -\frac{1}{2} \sum_{k=0}^d \log(2\pi \sigma_k) - \frac{1}{2} \sum_{k=0}^d \left(\frac{(\lambda_{\boldsymbol{\mu}}^c \log_{\boldsymbol{\mu}}^c(\mathbf{z}_k))^2}{\sigma_k} \right) + (d-1) \log \left(\frac{\sqrt{c} d_p^c(\boldsymbol{\mu}, \mathbf{z})}{\sinh(\sqrt{c} d_p^c(\boldsymbol{\mu}, \mathbf{z}))} \right) \\ &= -\frac{1}{2} \sum_{k=0}^d \left(\log(2\pi \sigma_k) + \frac{(\lambda_{\boldsymbol{\mu}}^c \log_{\boldsymbol{\mu}}^c(\mathbf{z}_k))^2}{\sigma_k} \right) + (d-1) \log \left(\frac{\sqrt{c} d_p^c(\boldsymbol{\mu}, \mathbf{z})}{\sinh(\sqrt{c} d_p^c(\boldsymbol{\mu}, \mathbf{z}))} \right) \end{aligned}$$

A decision was made early on in the development of the model to clip the $\log(\sigma^2)$ outputs of the encoder to $\{-10, 10\}$ to prevent instabilities. Furthermore, the variance for the prior distribution is chosen to be a fixed hyper-parameter. Hence, $\boldsymbol{\sigma}$ is never zero and so no considerations for numerical instability are required in the first term. The second term however involves $d_p^c(\boldsymbol{\mu}, \mathbf{z})$ which could well be zero should the embedding overlap exactly with the group mean, so numerical instability considerations are required here. Note that $\sinh(x) \approx x$ for small x . This implies that the second term is in fact undefined at $\boldsymbol{\mu} = \mathbf{z}$. To prevent this, small ϵ values can be added as follows:

$$(d-1) \log \left(\frac{\sqrt{c} d_p^c(\boldsymbol{\mu}, \mathbf{z}) + \epsilon}{\sinh(\sqrt{c} d_p^c(\boldsymbol{\mu}, \mathbf{z})) + \epsilon} \right)$$

So, the KL-divergence in the Poincaré ball model was approximated as:

$$L_{KL} = \frac{1}{|B|} \sum_{g \in G} \sum_{\mathbf{x} \in G_g} \log \mathcal{WN}_c(\mathbf{z} \mid \boldsymbol{\mu}_{\phi}(\mathbf{x}), \boldsymbol{\sigma}_{\phi}^2(\mathbf{x})) - \log \mathcal{WN}_c(\mathbf{z} \mid \boldsymbol{\mu}_g, \boldsymbol{\sigma}^2)$$

where,

$$\log \mathcal{WN}_c(\mathbf{z} \mid \boldsymbol{\mu}, \boldsymbol{\sigma}^2) = -\frac{1}{2} \sum_{k=0}^d \left(\log(2\pi \sigma_k) + \frac{(\lambda_{\boldsymbol{\mu}}^c \log_{\boldsymbol{\mu}}^c(\mathbf{z}_k))^2}{\sigma_k} \right) + (d-1) \log \left(\frac{\sqrt{c} d_p^c(\boldsymbol{\mu}, \mathbf{z}) + \epsilon}{\sinh(\sqrt{c} d_p^c(\boldsymbol{\mu}, \mathbf{z})) + \epsilon} \right)$$

It is important to note that a careful consideration must be taken when computing the wrapped

normal probability with a low variance and mean close to the edge of the ball. In this case, all samples are close to the edge of the ball and since distances grow exponentially, the distance between the sample $\mathbf{z}$ and mean $\boldsymbol{\mu}$ is increasingly large. Hence, $\log_{\boldsymbol{\mu}}^c(\mathbf{z})$ is often vary large. This leads to a significantly large first term in this expression which the model attempts to minimise further by reducing the variance since the influence of $\log(2\pi\sigma_k)$ is not great enough to prevent this.

Appendix B

The dataset was filtered and under-sampled prior to any train-test splits by the process described in section 3.1. A randomly selected 72% subset of samples was used for training, 20% used for testing, and 8% used for validation to inform early breaking conditions. To support stress testing, the highest magnitude sample was included in the test set, and the second highest in the validation set. To prevent data leakage, all samples from the same earthquake event were kept in the same subset.

The encoder and decoder neural networks were created using algorithm 1 with parameters $N = 1, \mathbf{P} = [1.0, 0.95, 0.9, 0.85]$ for the encoder, and $N = 2, \mathbf{P} = [1.0, 0.95, 0.9, 0.8, 0.7, 0.6, 0.5]$ for the decoder. The following hyper-parameter values were used in the final model: $d = 2, c = 1.0, \mathbf{W}^T = [30 \ 1 \ 50 \ 1000 \ 10000], \sigma_g^2 = 0.1, s = 2.0$.

The model was trained for a minimum of 300 epochs, maximum of 500 epochs, and the following early breaking condition: For either training loss or validation loss, for the maximum across all loss terms, if the average decline in loss over the past 10 epochs is less than 0.001 of the loss at the tenth previous epoch, break. This ensured all loss terms were sufficiently settled, regardless of their magnitude.

$$\max_{\ell \in \mathcal{L}} \frac{\text{mean}(\ell_{t-11:t-1} - \ell_{t-10:t})}{\ell_{t-11}} < 0.001.$$

Appendix C

In Poincaré ball geometry, distances grow exponentially away from the origin. If samples are embedded close to the edge of the ball, then this can lead to negative KL-divergence due to the approximation via probability densities. In such cases, it is highly plausible for the sample to be closer to the prior mean than the posterior mean (Eq. 13) such that $q_{\phi}(z | x) < p_g(z)$, particularly true if the posterior variance is significantly small.